

Lab 2 - Java Parallel Programming and Sorting Algorithms

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Task 1: Sequential Sort

We chose to implement MergeSort

Source files:

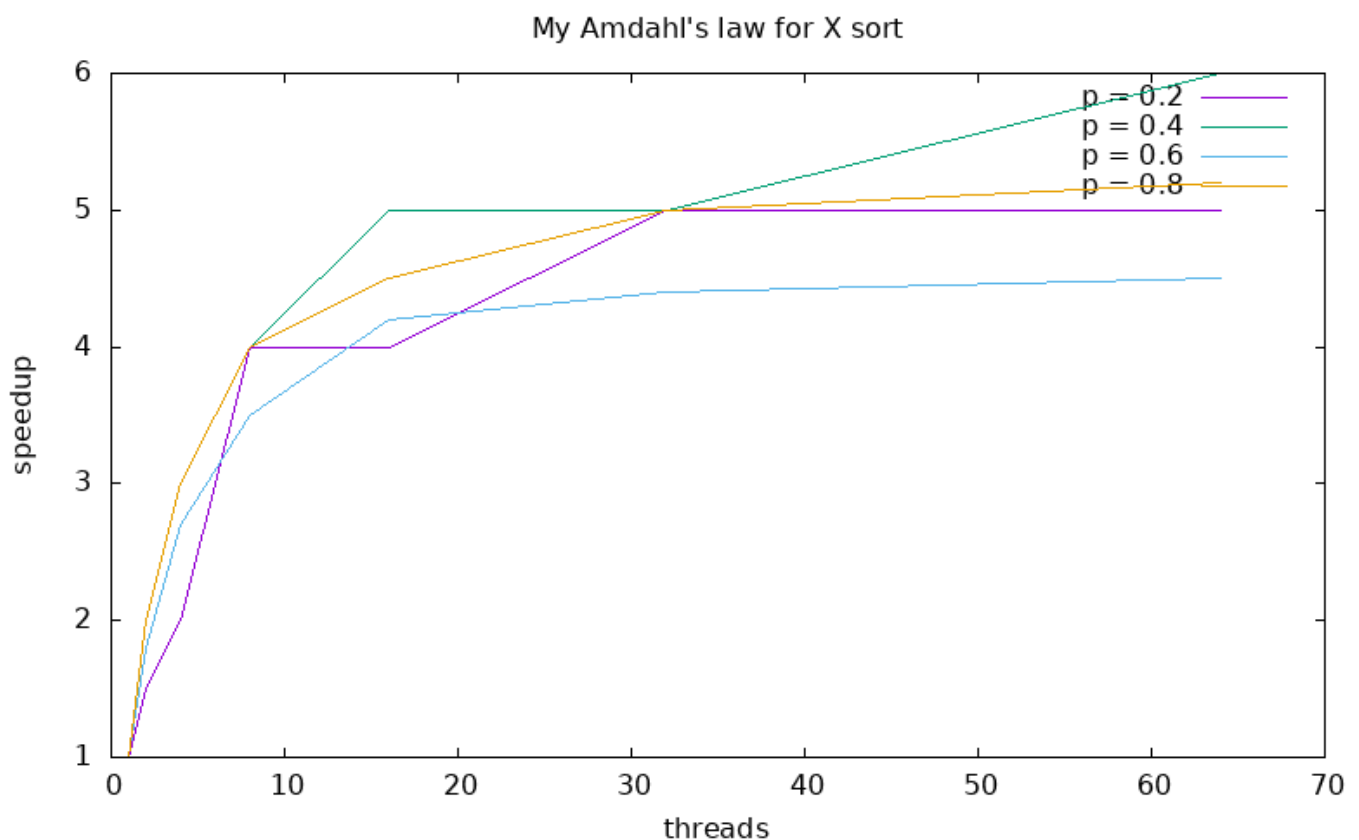
- `SequentialSort.java`

Task 2: Amdahl's Law

Classical Amdahl's law:

$$S(N) = \frac{1}{(1 - p) + \frac{p}{N}}. \quad (1)$$

Here is a plot of our version of Amdahl's law



We see that when $threads < 15$, the speedup increases sharply; beyond that point, adding more threads brings negligible returns.

- As the thread count N grows, recursion produces smaller subproblems.

- As thread increases, task scheduling, synchronization and cache effects cost more

Our model: Amdahl + overhead

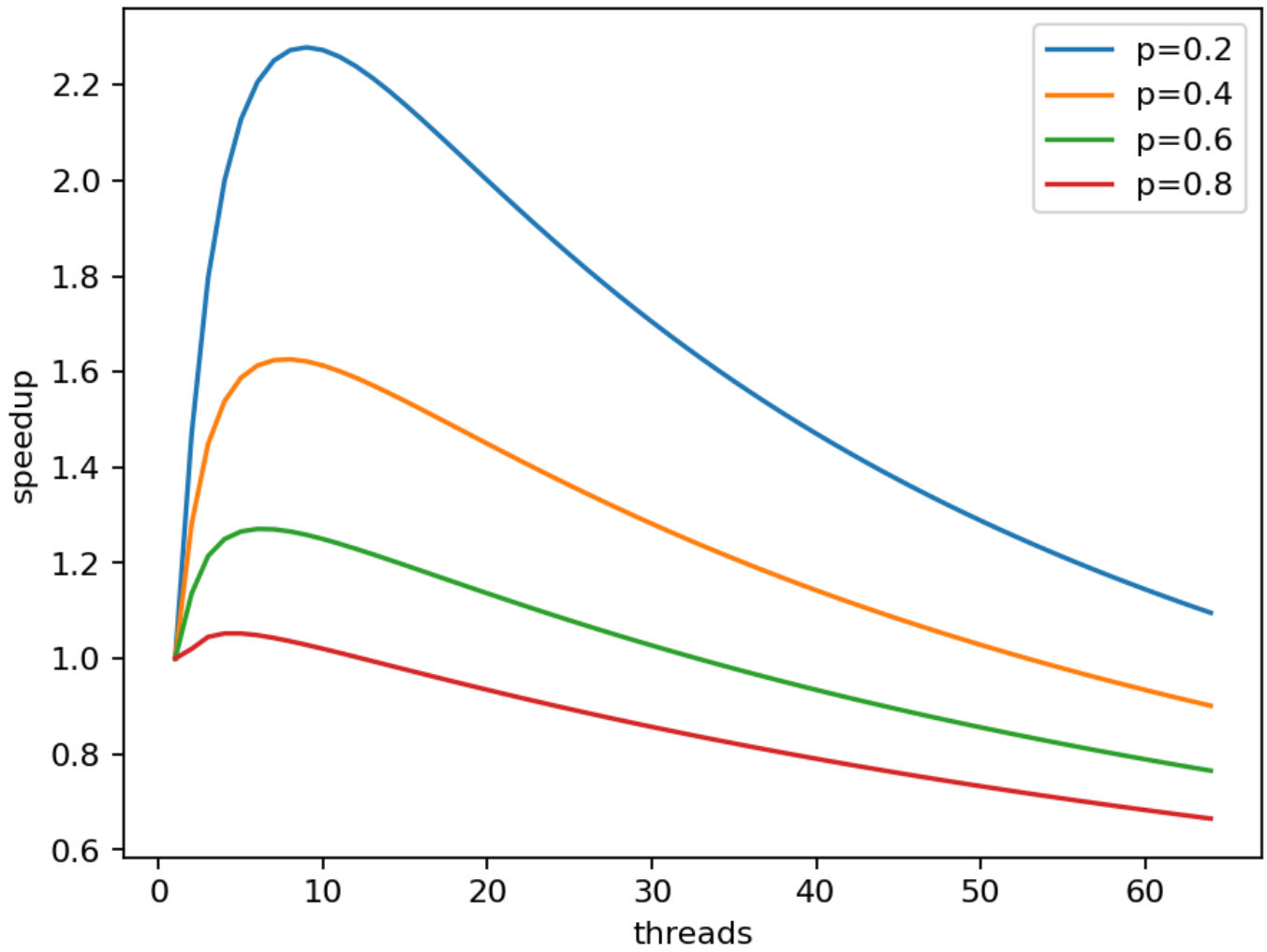
We add a linear overhead term to classical Amdahl's law:

$$S(N) = \frac{1}{(1 - p) + \frac{p}{N} + \delta(N)} \quad (2)$$

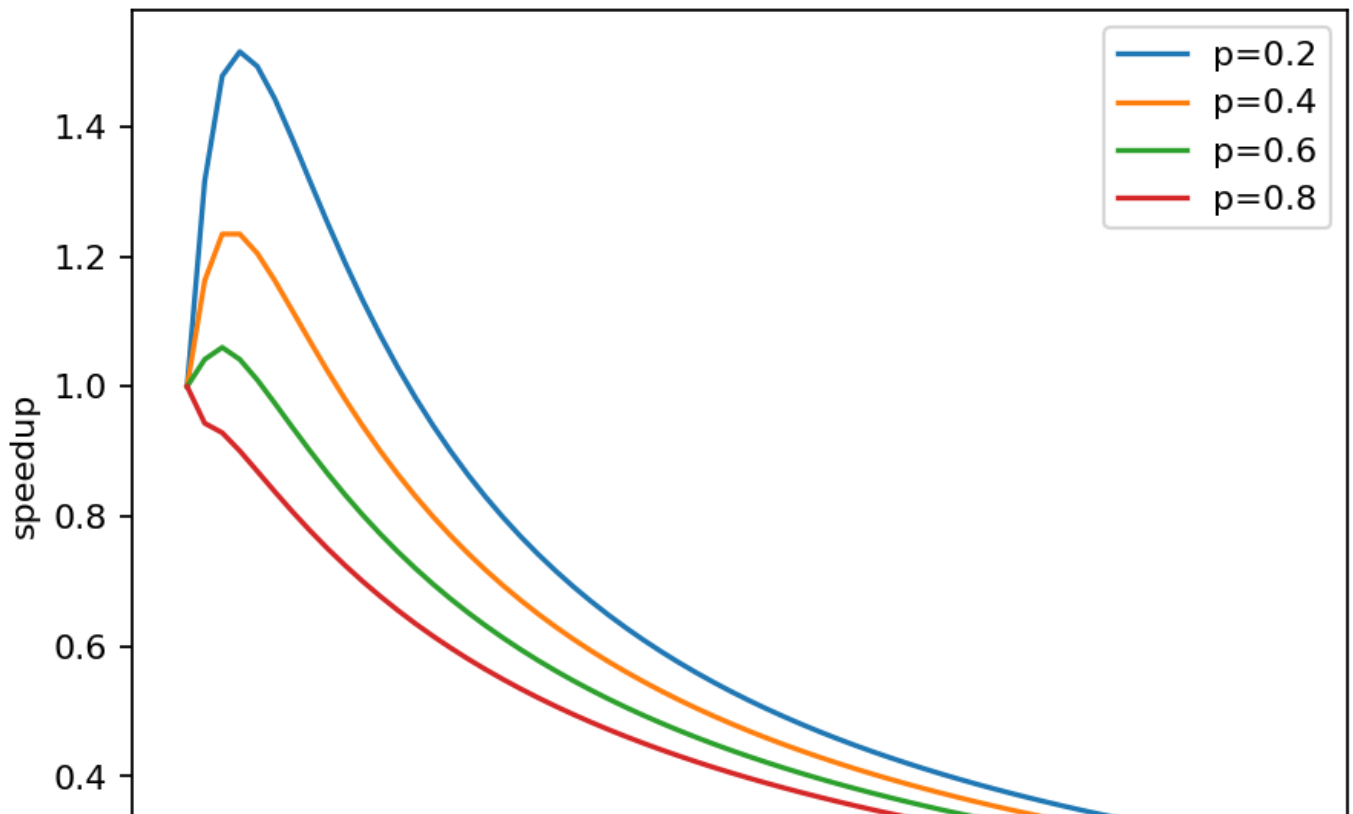
$$\delta(N) = a + bN \quad (N > 1), \quad \delta(1) = 0. \quad (3)$$

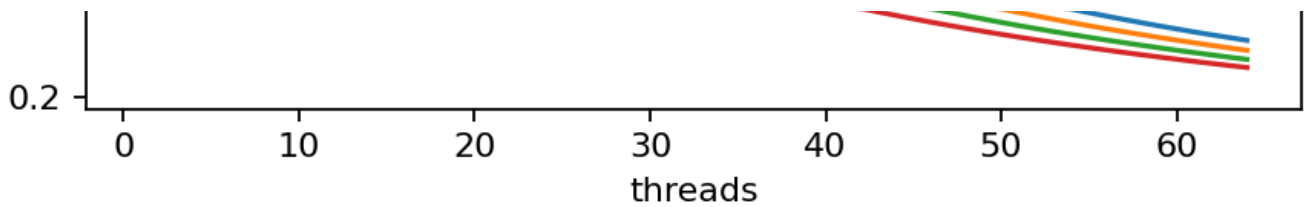
Classical Amdahl assumes perfect division of work and negligible overhead, which overestimates speedup. By adding $\delta(N)$, we capture N -dependent costs (scheduling, synchronization, cache effects), so the curves visualized realistically as N grows.

Overhead model ($\Delta(N)=a+bN$, $a=0.06$, $b=0.01$)



Overhead model ($\Delta(N)=a+bN$, $a=0.06$, $b=0.05$)





Task 3: ExecutorServiceSort

Source files:

- `ExecutorServiceSort.java`

We implement a fixed thread pool with daemon workers to ensure the JVM exits cleanly after measurements, stop parallel recursion when the depth limit $d = \lfloor \log_2 Thread \rfloor$ is reached. This keeps the merge logic identical to the sequential version and prevents task explosion.

Task 4: ForkJoinPoolSort

Source files:

- `ForkJoinPoolSort.java`

We implement a parallel merge sort with a `ForkJoinPool` per subrange. Each task splits at the midpoint, runs both halves in parallel via `invokeAll`, and merges in the parent thread after the joins; stop parallel recursion when the depth limit $d = \lfloor \log_2 Thread \rfloor$ is reached.

Task 5: ParallelStreamSort

Source files:

- `ParallelStreamSort.java`

We first use

```
ForkJoinPool pool = new ForkJoinPool(threads);
```

to create a thread pool.

Then, we use `Arrays.stream` to have an stream object of the int array. We use `.parallel()` to have a parallel stream of the int array. We use `.sorted()` to sort the stream and use `.toArray()` to convert the result to array.

We use

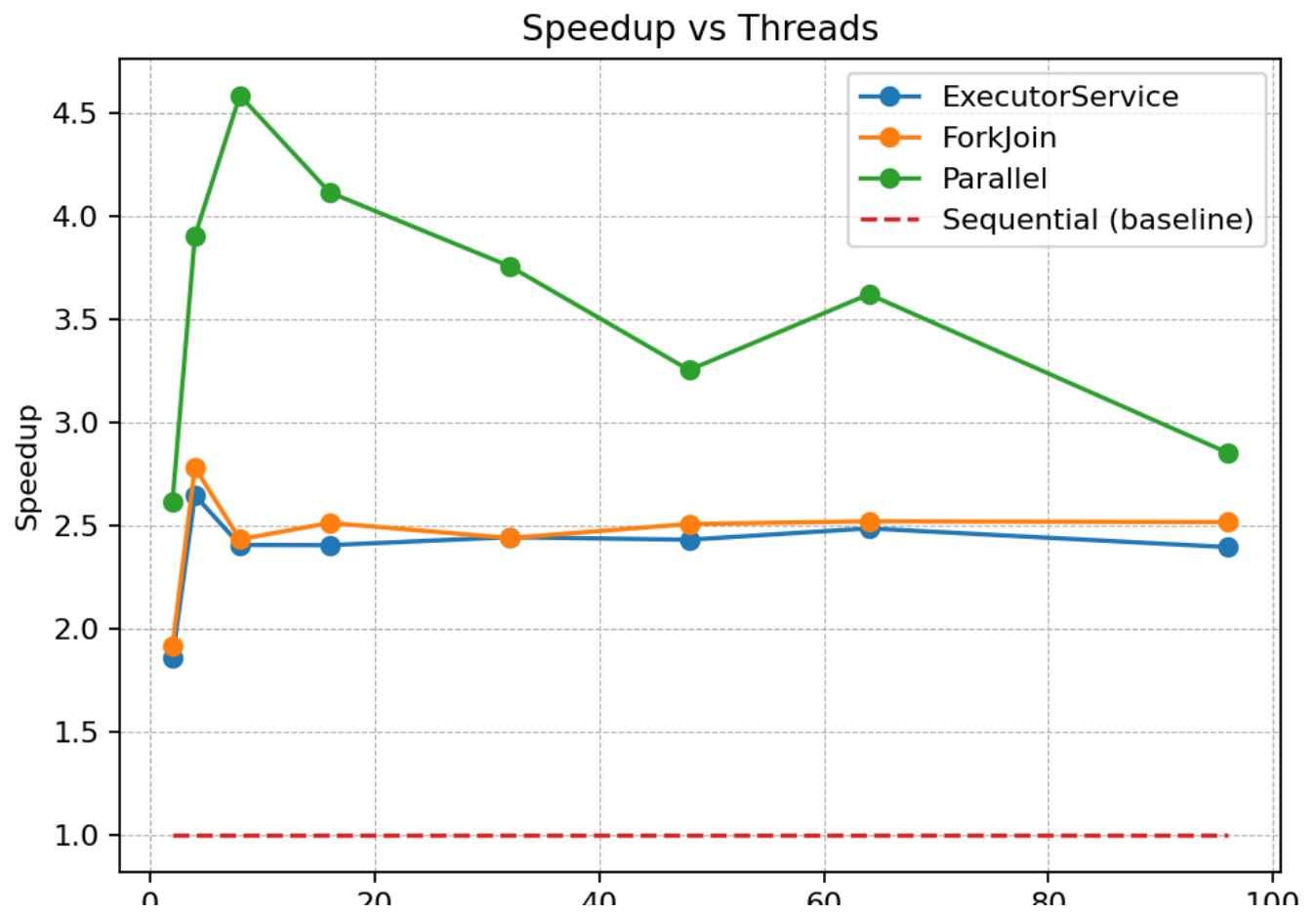
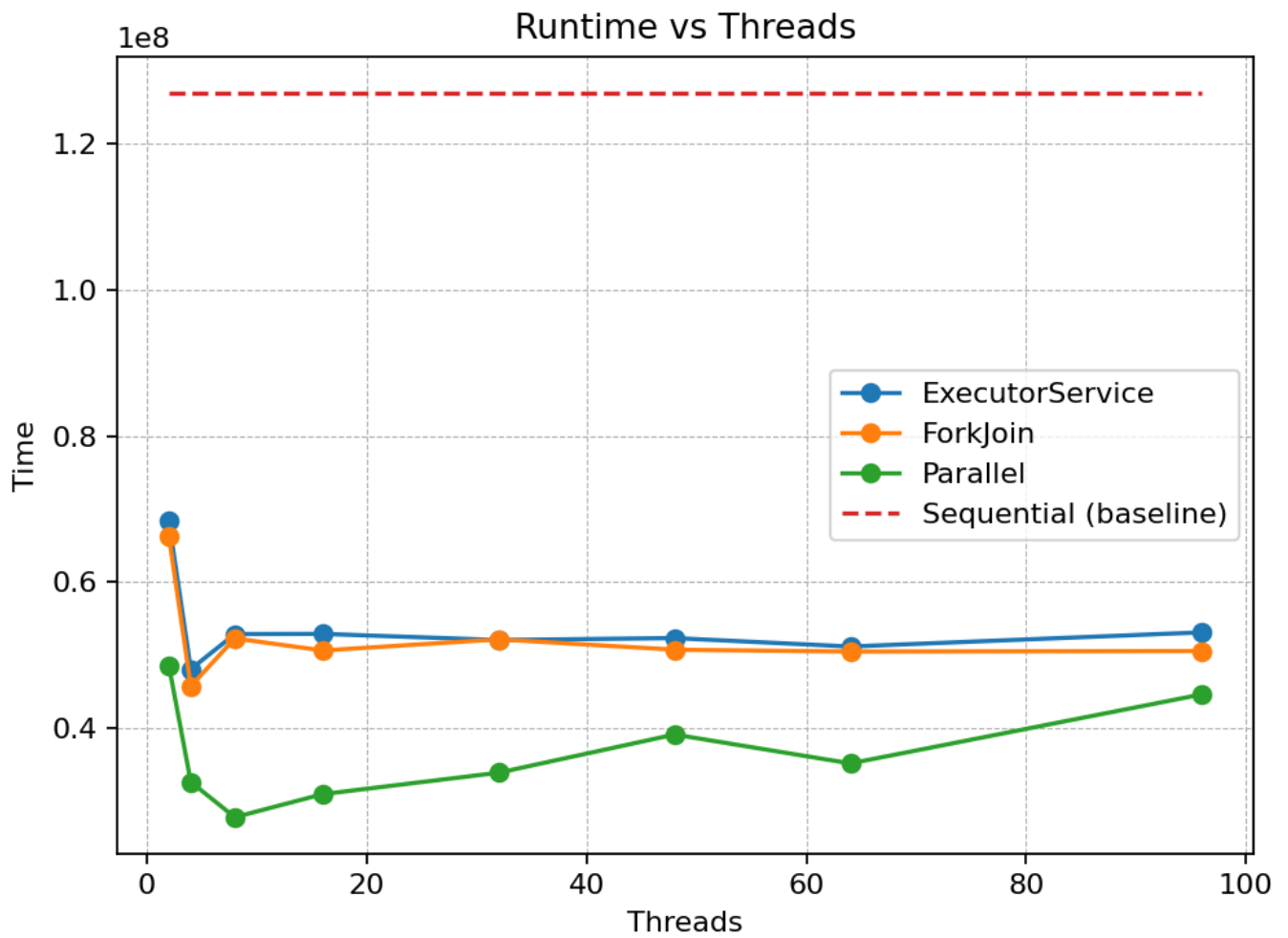
```
System.arraycopy(temp, 0, arr, 0, arr.length);
```

to copy the tempt result to `arr`

Task 6: Performance measurements with PDC

Out parameters:

- `SIZE=1000000`
- `WARMUP=10`
- `MEASURE=1000`
- `SEED=42`



0

20

40

60

80

100

Threads

Are the performance gains/drops between different numbers of threads what you expected?

Yes. Speed improves from 2→8→16 threads, then flattens as thread-management and memory overheads dominate.

What implementation ran the fastest/slowest? Was this in accordance with your expectations?

Fastest is `ParallelStream`; slowest is `Sequential`. `ExecutorService` and `ForkJoin` are close, which is expected since they share the same merge and depth cap and pay extra scheduling overhead compared to `ParallelStream`.

What method was the easiest to implement?

`ParallelStream`, no manual recursion, merge code, or tuning; just a one-line parallel stream.

What method do you prefer?

`ParallelStream`, because it is easy and performs best.